

Abstract:

Medical image segmentation is a fundamental task in intelligent diagnosis, lesion quantification, and treatment planning. However, existing supervised methods usually rely on a large amount of high-quality pixel-level annotations, which leads to high training costs and long data preparation cycles, thereby limiting their applicability in real clinical scenarios. This challenge is particularly prominent in settings involving rare diseases, primary healthcare institutions, and cross-center datasets, where annotation scarcity has become a major bottleneck for the deployment of medical image segmentation models. Meanwhile, targets in medical images often exhibit substantial scale variation and morphological diversity. A single image may simultaneously contain large anatomical structures, ambiguous boundary regions, and small-volume lesions. Traditional single-scale or weak multi-scale methods often struggle to balance local detail representation and global contextual modeling, which in turn affects segmentation accuracy and robustness. To address these challenges, this study focuses on low-label medical image segmentation and cross-scale representation learning, aiming to explore effective approaches for achieving accurate and generalizable segmentation under limited annotation conditions.

This study plans to build a cross-scale medical image segmentation framework for low-label settings based on a U-Mamba-like hybrid architecture, combining the local modeling capability of convolutional networks with the long-range dependency modeling capability of state space models. The feasibility of this topic is reflected in three aspects. First, low-label medical image segmentation addresses a clear practical demand and has stable application scenarios, making the research problem well-defined and the experimental objectives explicit. Second, existing U-shaped encoder-decoder architectures, Mamba/SSM modules, and self-supervised or masked pretraining strategies have provided a relatively mature technical foundation for method design and experimental validation. Third, the proposed study can be conducted progressively on public medical segmentation datasets by first reproducing baseline models under low-label settings and then gradually incorporating cross-scale modeling and pretraining-enhanced modules, resulting in a clear technical route and controllable implementation risk.

The innovation of this study is mainly reflected in the following aspects. First, to address the insufficient supervision in low-label settings, a feature enhancement and representation learning mechanism tailored to the hybrid CNN-SSM architecture will be designed to improve the utilization efficiency of limited labeled samples. Second, considering the significant scale variation of medical targets, cross-scale feature interaction and multi-level semantic fusion strategies will be introduced to enhance the recognition of small lesions, elongated structures, and complex boundary regions. Third, instead of treating “low-label” and “cross-scale” as two separate issues, this study investigates the stable learning of multi-scale representations under limited supervision within a unified framework, thereby improving the robustness and generalization ability of the model in few-sample, cross-dataset, and structurally complex scenarios. It is expected that this work will reduce the dependence on dense annotations while improving the overall segmentation performance on multi-scale targets, thus providing theoretical support and methodological guidance for the development of more clinically applicable intelligent medical image segmentation approaches.